Lee Kong Chian School of Business, Singapore Management University

Marketing Management

Week 8 · Managing Resource Trade-Offs

Marketing Principle #4: All Resources Are Limited

Yixing Chen

Agenda

01 **Introduction**
Why limited and interdependent resources create recurring trade-offs

02 **Attribution and response models**
Heuristics, experiments, response functions, calibration, and optimization

03 **Marketing metrics**
Intermediate measures, financial metrics, and dashboards

All Resources Are Limited

- The final and perennial issue facing managers is that all resources are limited and often interdependent
  - → Managing resources optimally is critical, because marketing resources provide the primary action levers that firms can use to implement what they have learned from the previous three Marketing Principles
  - → The previous 3 Mkt. Principles all involve tradeoffs inherent in making strategic decisions (STP, AER, BOR) while also providing input for the 4th Mkt. Principle, which determines the annual budget to specific marketing actions (plan, budgets, and metrics)

- Specifically, we need to balance marketing resources across:
  - → **Customer segments**: targeting & positioning (STP)
  - → **Lifecycle stages**: acquisition, expansion, and retention (AER)
  - → **Customer equity and marketing mix elements**: brand, offering, and relationships (BOR, 7Ps)
  - → **Other:** SCAs now and for future, product categories, R&D projects, geographies, sales territories, etc.

-

Exercise on Resource Tradeoffs

- Everyone take a few minutes to describe how your firm allocates the sales and marketing/R&D budget $ across the following areas:
  - → Customers (individuals, segments, geographic)

_____

  - → Stages (acquisition, expansion, and retention)

_____

  - → Marketing mix (brand, offering, relationships, or 7Ps)

_____

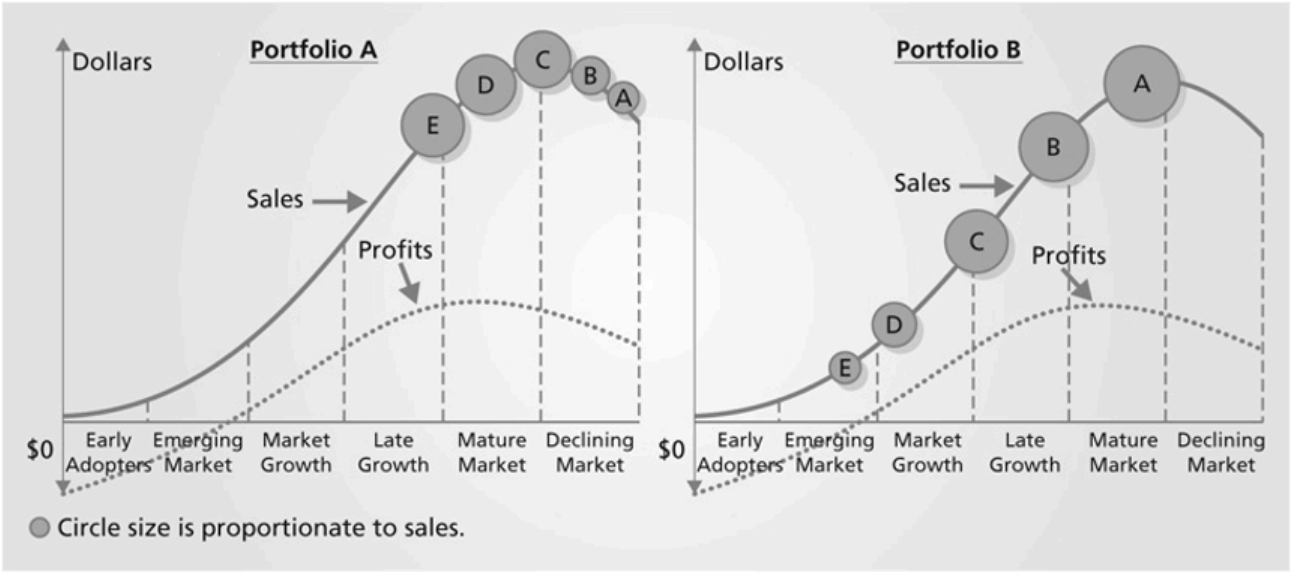
- Does this results in any misallocations, describe?

_____

- Why do these misallocations occur?

_____

Firms Often Make Poor Allocation Decisions



- How do these portfolios differ?
- Which is stronger?
- Why would this happen?

Source of Needing to Change Resource Tradeoffs

- **Resource slack** refers to the "potentially utilizable resources a firm possesses that it could divert or redeploy to achieve organizational goals"
- Changes in customers' needs
- Changes in the lifecycle stage of a firm's products
- Changes in the product market landscape, due to entries and exits by competitors
- Changes in the effectiveness of marketing activities
-
-
-

Example: Smith's Snackfood Company (UK / Australia)

- Smith's Snackfood Company, is a British-Australian company producing snack foods
- The company records all the inputs and outputs throughout a product's lifecycle from pre-far preparations, on-farm processes to post-farm transportations
- The data showed that one of the significant trade-offs during the packaging and processing stage for corn chips is between resource usage and emission of greenhouse gases
- This assessment enables Smith's Snackfood to make more informed decisions on its resource allocations across the product lifecycle stage.

Sources of Resource Trade-Offs

Sources of Resource Trade-Offs, Con't.

All Resources Are Limited: A Fundamental Assumption of Marketing Strategy

- The recognition that all resources are limited and that an effective marketing strategy must manage ever-present resource trade-offs is the fourth and final Marketing Principles (MP#4)
- The approaches for dealing with the three previous Marketing Principles in many ways help create the MP#4
- All resources are constrained; even if a firm's existing resource allocation policies appear effective, rapid and often unexpected changes in the legal, economic, technological, or innovation landscape demand constant vigilance
- Firms thus need ways to identify misallocations and adjust spending levels quickly, in response to each new situation
- A firm's marketing strategy must account for resource trade-offs, or else its ability to execute the initial three First Principles of marketing strategy, and thus its overall marketing strategy, will suffer

Need For Resource Tradeoffs Across First Principles

- Selecting the **who and what to "sell"** across all potential customers in the market
  - → **Approach:** segment and target like customers (STP)
  - → **Decision criteria & allocation process**: Attractiveness and competitive strength using GE matrix
  - → **Improvements:** Smaller segments and CLV
- Selecting **who and what to "do"** across portfolio of your existing customers
  - → **Approach:** segment across stages (AER)
  - → **Decision criteria & allocation process**: CLV of customer personas
  - → **Improvements:** Individual customer-level analysis, HMM for dynamic segmentation
  - →
-
-

Need For Resource Tradeoffs Across First Principles (con't)

- Deciding **how to use brand, offering, and relationships** to achieve positioning objectives from STP and AER
  - → **Approach:** customer equity stack (BOR)
  - → **Decision criteria & allocation process (within domain)**: elasticities, partworth utilities, and CLV from choice models, conjoint analysis, experiments, and regression
  - → **Improvements:** Multivariate (simultaneous) models
- Determining **annual budgets, plans, and metrics to execute** BOR marketing strategies and support positionings
  - → **Approach:** act, measure metric, evaluate, adjust (iterative)
  - → **Decision criteria & allocation process**: MROI, CLV, elasticities and partworth utilities using choice models, conjoint analysis, response models, and Solver
  - → **Improvements:** More refined metrics and more inclusive response/optimization models (variables and time)
-
-

Agenda

01 **Introduction**
Why limited and interdependent resources create recurring trade-offs

02 **Attribution and response models**
Heuristics, experiments, response functions, calibration, and optimization

03 **Marketing metrics**
Intermediate measures, financial metrics, and dashboards

Evolution of Approaches for Managing Resource Trade-Offs

1

Heuristic Era — **Information Available for Evaluating Resource Trade-Offs** — **Data Era**

Very little hard data

Detailed data on individual consumers over time

(a) Heuristic Era: Anchoring and Adjusting Approach

Anchoring

Marketing Resources Advertising, sales force, etc.

(Not evaluated by managers)

Marketing Outcomes Market share, sales, customer loyalty, etc.

(Feedback from past outcomes)

Adjusting

(b) Data Era: Attribution Approach

Marketing Resources Advertising, sales force, etc.

Attribution Model Experiments, response models, etc.

Marketing Outcomes Market share, sales, customer loyalty, etc.

(Feedback from past outcomes)

Adjusting

Trends that Have Changed How Marketers Allocate Resources

Poorly developed	**Information Technology Infrastructures**	Well developed
Mostly qualitative and ad hoc data	**Volume of Data**	Longitudinal, customer-level data over time
Uninformed customers	**Customer Sophistication**	Highly informed customers
Trust in marketing effectiveness	**Proof of Marketing Effectiveness**	High demands of mathematical proof of effectiveness
Local markets	**Globalization**	Highly competitive, global markets

Evolution of Approaches for Managing Resource Trade-Offs

- The evolution of approaches for managing resource allocation suggests two main and overlapping eras
- **Heuristics Era** – firms constantly decide how to allocate resources across different customer segments, different customer stages, different offerings, different regions, and different marketing communication formats
  - → No hard data about resource allocation options
  - → Managers used simple rules of thumb for allocating resources
- **Data Era** – firms started using historical data that revealed the link between their past resource trade-off decisions and outcomes, such that they could determine the actual effects of certain resources on specific outcomes
  - → Scientific approaches, based on data and empirical models, reveal whether firm should continue its level of resource commitments or adjust them

Example: Optimal Strategy for Advertisements (China)

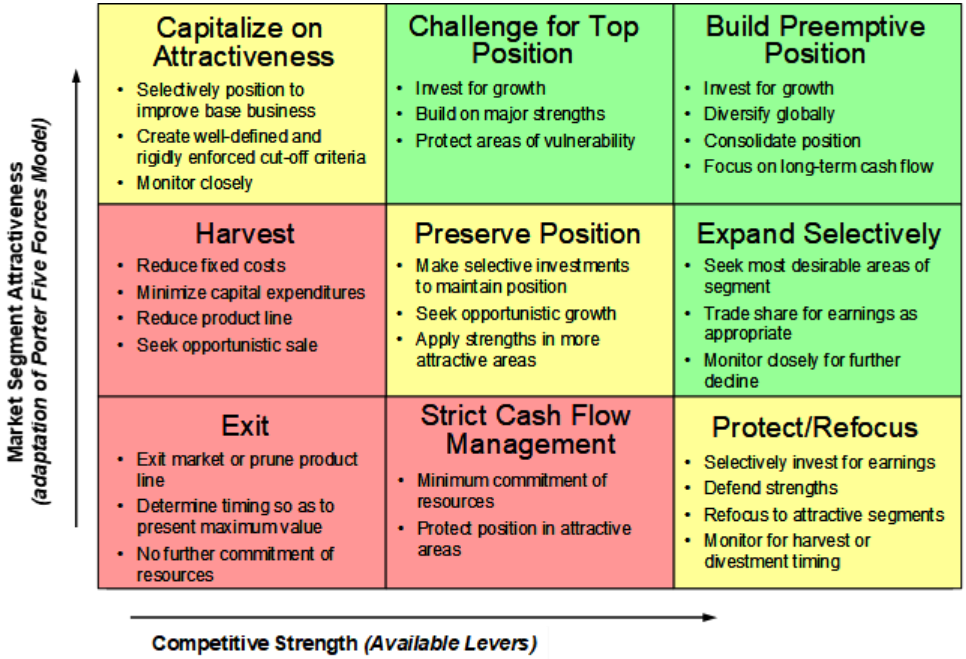
- To discern whether and when to target consumers with mobile advertising, a large Chinese mobile phone provider conducted a large-scale study in which it sent 4,400 mobile users a promotional text between 8 am and 8 pm
- The consumers responded directly to the offer, paying using their phones
- With the resulting information about when and how consumers responded to the advertisements, the firm was able to determine its optimal strategy, which was to target customers between 10 am and 12 pm, and then again between 2 pm and 4 pm
- By doing so, it could prompt average purchase rates that were twice what the firm had been achieving

No "Perfect System" for Allocating Resources

- **Problem:** Needs to be *simple* enough so it is transparent to managers (believed and understood) while being *sophisticated* enough to account for the interdependencies across multiple time-varying inputs each with different elasticities

- **"Best" method**
  - → Iterative approach which attempts to make optimal allocations across nested levels of decisions
  - → Each level can be understood (bounded rationality of managers) while allocation decision criteria and approach can be selected based on characteristics of that level
  - → Can learn and improve at each level overtime (short feedback loop with targeted metrics at each level; build specific expertise)
  - → Attempt to account for cross-level effects on a case-by-case basis
  - →

GE Portfolio Analysis is Classic Approach for Making Resource Tradeoff Decisions



- What are the pros and cons of this approach?
- How does it account for AER, BOR, and SCA tradeoffs?

Anchoring – Adjusting Heuristics Approach

- **Anchoring and adjustment heuristics** are widely used, though the exact anchoring rules vary in practice:
  - → In the percentage of sales method, marketing resources reflect the sales revenue earned from the focal product
  - → With the percentage of profits method, the resources dedicated to marketing instead vary with the profits earned by the product in previous periods
  - → Managers who adopt the historical method simply set their present resource allocations to a level very close to the previous year's spending
  - → Finally, the competitive parity method implies that managers set resource allocation levels to match those of their competitors

Attribution Approach

- Specifying an attribution model, such that the firm learns the exact dollar impact that a small resource increase will exert, also can provide answers to two more questions:
  - → What is the relative dollar value impact of a marketing investment?
  - → What is the profit-maximizing level of investment?
- By using an attribution model, managers thus can allocate resources to optimize their desired outcome, as well as avoid waste or reliance on arbitrary heuristics
- In turn, rather than reactive resource allocation strategies, managers can implement proactive ones

Experimental-Based Attribution

- Firms operate in environments in which various factors operate simultaneously
- Managers still must make constant, rapid decisions about whether to commit resources and how much to commit
- The experimental attribution approach involves an intervention, outcome, design of the experimental condition, and a control condition
- With this approach, all other factors (at least those under the firm's control) that can influence sales are purposefully kept constant
- This method offers many advantages over an anchoring–adjustment approach

Experimental-Based Attribution (con't)

- Deciding which factors to test is critical, and experiments can quickly grow very complex
- Another element that defines the success of an experimental approach is its reliability, that is, its internal and external validity
- Internal validity means that the experiment is well designed, and it reflects three key criteria:
  - → The cause should precede the effect in time, known as **temporal precedence check**
  - → The cause and the effect must be related, according to a **covariation check**
  - → No plausible alternative explanations exist for the observed outcome, which can be determined with a **non-spuriousness check**

Components of Experimental Attribution

Response Model Attribution

- A response model is a statistical model that captures the relationship between past marketing resources and past outcomes
- The underlying philosophy is that historical data contain insightful information about whether and how much marketing resources truly increase outcomes, which is useful to know when deciding on future marketing actions
- With the basic—and often reasonable—assumption that past outcomes relate to future outcomes, this approach leverages the past data to isolate the relationship between marketing resources and performance
- Response models also offer many advantages, in terms of their flexibility and usefulness

Example: Samsung Electronics (South Korea)

- In 1999, Samsung Electronics needed to allocate its corporate budget of $1 billion across 14 products, sold in more than 200 countries, with the goal of improving the returns on its marketing spending
- Before the reallocation, Samsung used an anchoring and adjustment method, allocating resources to products and countries in proportion to the sizes of their markets
- When it adopted a response model approach, Samsung learned that it had overinvested in North America and Russia. It reduced spending in those regions and invested more in Europe and China
- By 2002, Samsung already had achieved significant market share gains in these countries, increased its brand value by 30 percent (to $8.3 billion), and grew its net income to $5.9 billion
-

Response Models

A response model estimates how marketing inputs relate to customer or financial outcomes. Managers use the estimated response function to compare investments and improve resource allocation.

When to use it

- Estimate the shape of the relationship between marketing effort and performance.
- Compare the impact of alternative marketing-mix investments.
- Account for competitor actions and other market conditions.

Marketing inputs → Response function → Expected outcomes → Allocation decision

How Response Models Work

01 Historical variation reveals how outcomes change when marketing inputs change.

02 Functional form captures diminishing returns, delays, or saturation.

03 Elasticities put different marketing activities on a comparable scale.

04 Competitor spending and market conditions enter as controls.

Illustrative log-log model

$$\ln(Y) = \beta_0 + \beta_1 \ln(X_1) + \beta_2 \ln(X_2) + \beta_3 \ln(Z_1) + \varepsilon$$

Y is the outcome, **X_1 and X_2** are the firm's marketing inputs, and **Z_1** captures competitive effort.

Each β coefficient is an elasticity: the expected percentage change in Y for a 1% change in the corresponding input.

Example: XYZ Newspaper

Decision problem

XYZ must divide a limited budget between newsroom quality and the advertising sales force during a difficult economic period.

Data

Ten years of monthly revenue, newsroom investment, sales-force investment, and competitor investment.

Model

A log-log response model estimates the elasticity of revenue with respect to each investment.

Variable	Elasticity	p-value
Newsroom investment	0.36	0.03
Sales-force investment	0.24	0.02
Competitor newsroom investment	−0.12	0.01
Competitor sales-force investment	−0.08	0.02

XYZ Newspaper: Interpretation and Decision

0.36
Newsroom elasticity

1.5×

0.24
Sales-force elasticity

Recommended reallocation

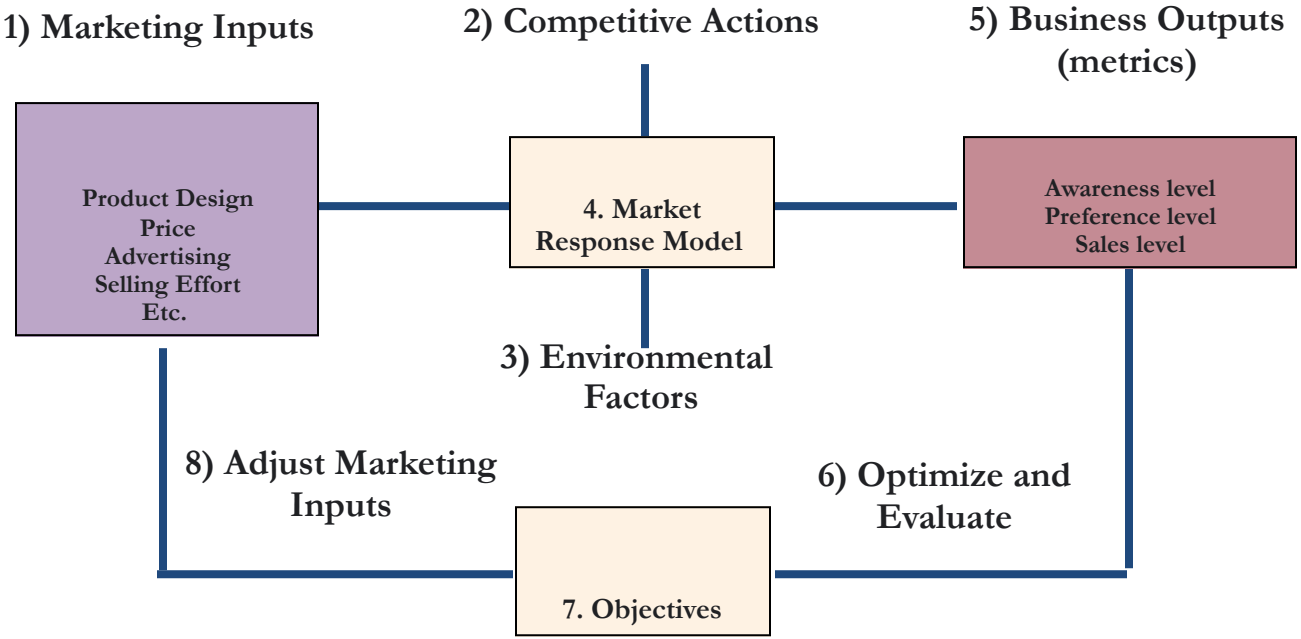
- Shift part of the budget toward newsroom quality because its estimated revenue elasticity is higher.
- Retain sales-force investment because its estimated effect remains positive and statistically significant.
- Monitor competitor newsroom investment, which has the larger negative association with XYZ revenue.

The model supports a reallocation decision. A field experiment should test whether the predicted lift is causal and transferable.

Resource Allocation Using Market Reponses Models: Pros and Cons

- Classic Approaches to Sales Forecasts and Marketing Budgets
  - → Percent of Sales Method
  - → Industry Benchmarks
  - → Last Period +/- Reward and Punishment
  - → "Squeaky Wheel" Model
  - → "Spread the Butter" Model
- Response Model
  - → Builds a "black box" model based on past response to changes in inputs
  - → Once model is built then optimizes for objective (Solver)
  - → In essence it is just curve fitting, which assumes past is good predictor of future

A Response Model System Has Eight Key "Parts"



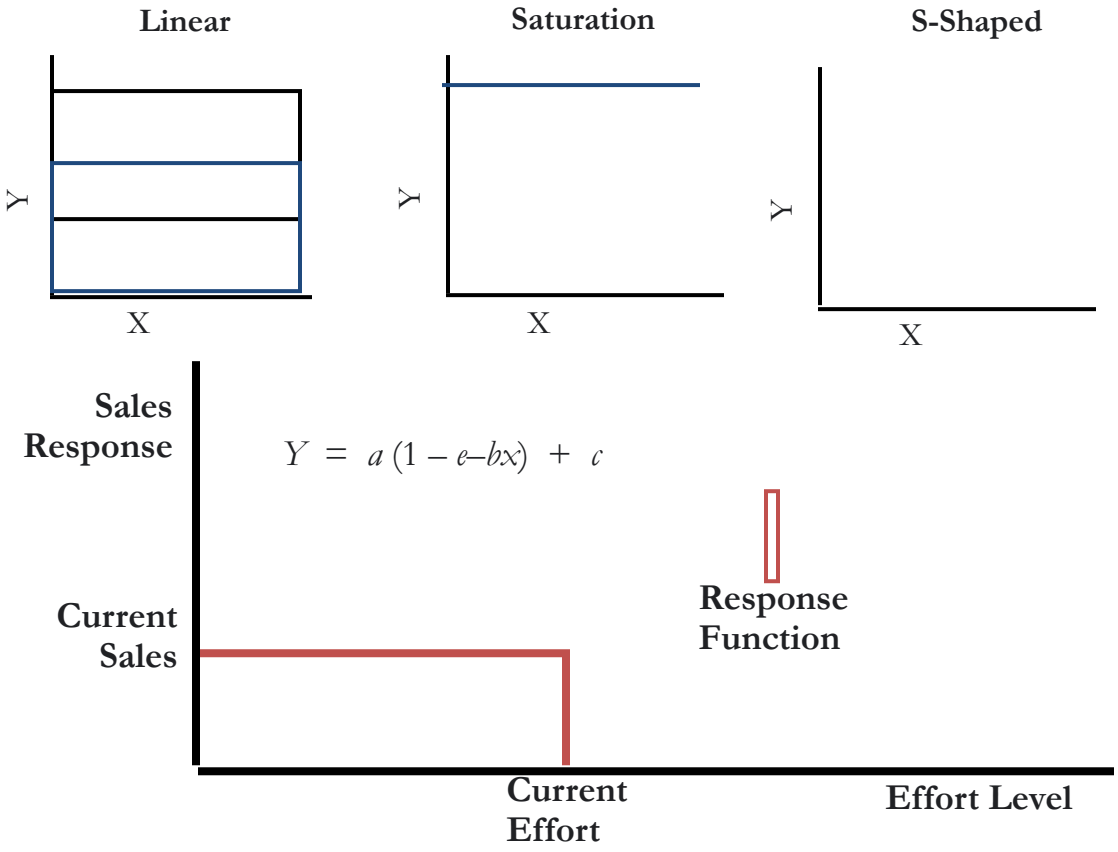
Steps to Building a Response Model

- Determine relevant inputs, outputs, and objectives
  - → Marketing actions (price, promotion, sales coverage)
  - → Competitive actions (price, new products, discounts)
  - → Environmental factors (government, trade barriers)
  - → Business outcomes/metrics (sales level, awareness)
  - → Objective (optimize sales or profits, advertising or sales coverage)
- Determine the "form" and "calibrate" the response model
  - → **Form** (i.e., type of equation)**:** linear, saturation, hysteresis, etc.
  - → **Calibrate:** estimate parameters using experiments, manager judgments, past data (need variance across inputs)
- Estimate response model and evaluate (vs. objectives)
- Optimize (Solver) to adjust marketing inputs
- Design field experiment and test
- Rollout or adjust model and redo process
-

Select Response Model's Functional Form That Captures Reality

- **Base level**: "sales" with no input
- **Linear**: constant relationship between input and output (infrequent)
- **Delayed response**: lag between input and output (common)
- **Saturation**: curve flattens at some higher level (diminishing returns, common)
- **S shape**: slow to start and reach saturation (good starting point)
- **Hysteresis**: slow to decay after input removed

Examples: Response Model's Functional Form



Calibration is the Process for Estimating Parameters in Response Model

- Estimate from historical data
  - → Data from CRM system
  - → Must be sufficient variability in the data to allow for reliable estimation (across regions, overtime)
- Conduct experiments
  - → Fairly easy in online settings
  - → P&G does this for advertising for most new product launches (3 cities)
- Elicit managerial judgment
  - → Quick and helps build consensus (combine with others)
  - → Encourages systematic thinking
  - → A series of questions on what would happen when inputs change

Preprogrammed Response Models (many included in ME)

- ADBUDG: market share response model for advertising spend, which builds in many known advertising phenomena
  - → Floor to share w/o advertising
  - → Ceiling where advertising spend has no effect
  - → Need some advertising rate to maintain share
  - → Built in typical delays and carryover
- Sales force allocation models
  - → GEOLINE: allocation of salespeople over territory
  - → CALLPLAN: allocation of sales calls over customers, products
  - → MDZ: allocation of compensation over salespeople
- Many other response models account for multiple types of marketing inputs simultaneously
  - → Interaction among advertising, salespeople, and promotion $
  - → Consultants specialize in each model and across industries
  - → Build your own model for key ongoing allocation decisions

Solver: An Optimization Program in Excel

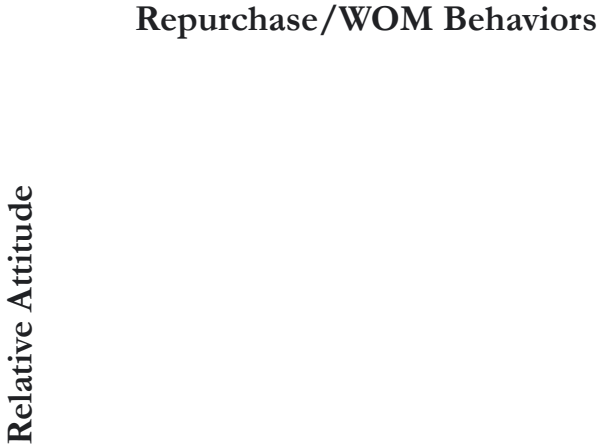
- Add-in for Excel that "solves" linear and non linear functions for optimal point
- Plug in equation for response function to optimize (set min and max limits)
- Set any constraints (e.g., advertising spending > 0)
- Solver gives optimum level of input(s) "assuming" response function captures relationship between inputs and outputs

"If you can't measure it, you can't manage it!" The Balanced Scorecard

- Managers need intermediate metrics to capture the effects of their decisions in order to adjust actions
  - → Short feedback loop, forward-looking, fast response
  - → Scope of metric should align to domain of decision
- Common marketing metrics:
  - → Customer satisfaction (product focused, forward-looking)
  - → Service quality (experience and relationship focused)
  - → Brand awareness (promotion focused, fast response)
  - → Brand strength and quality (promotion, price, and product focused, slow response/high inertia)
  - → Customer retention (aggregate, backward-looking)
  - → **Loyalty** (probably best forward-looking aggregate measure of marketing's effect)
-
-

However, Some Measures of Loyalty are Misleading

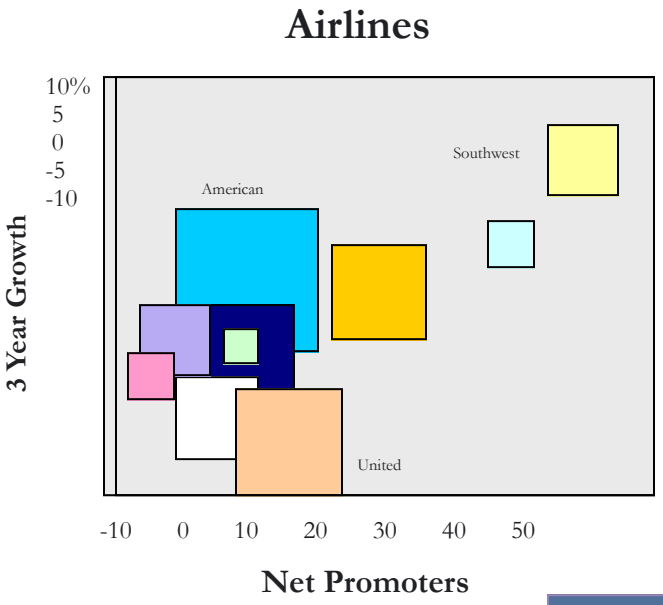
- **Loyalty:** attitudes and behaviors that signal a motivation to maintain a relationship
  - → **Attitudes**: captures beliefs and feelings relative to competitive options
  - → **Behaviors**: captures actual actions

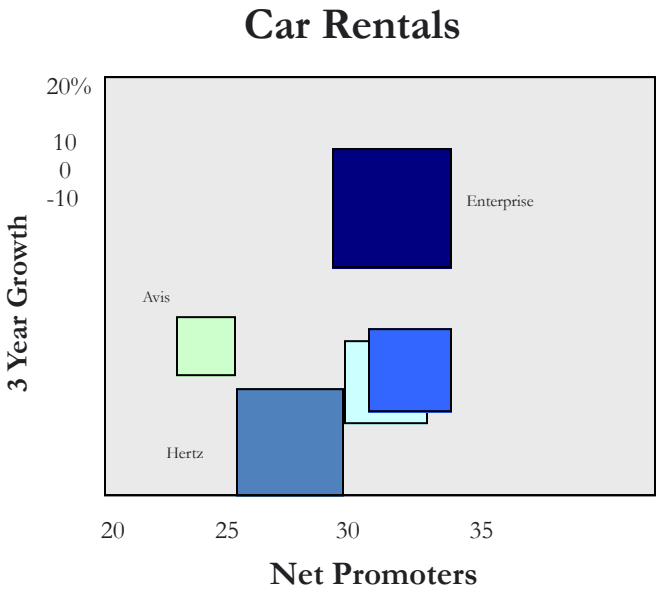


Net-Promoter Score (NPS): Popular Loyalty Measure

- **Net-promoters:** % of customers who are promoters - % who are detractors (i.e., removes people in zone of tolerance)
- Promoters answer 9 or 10 and distractor 0 to 6 to question, "How likely is it that you would recommend [firm] to a friend or colleague?"
- Scores are linked to profitable growth
- Scores of 75% and above are world class
- Mixed support in academic research
- **Pros**: simple and good "single item" measure
- **Cons**: not very diagnostic, may not be true loyalty since no actual behaviors

Growth by Word of Mouth





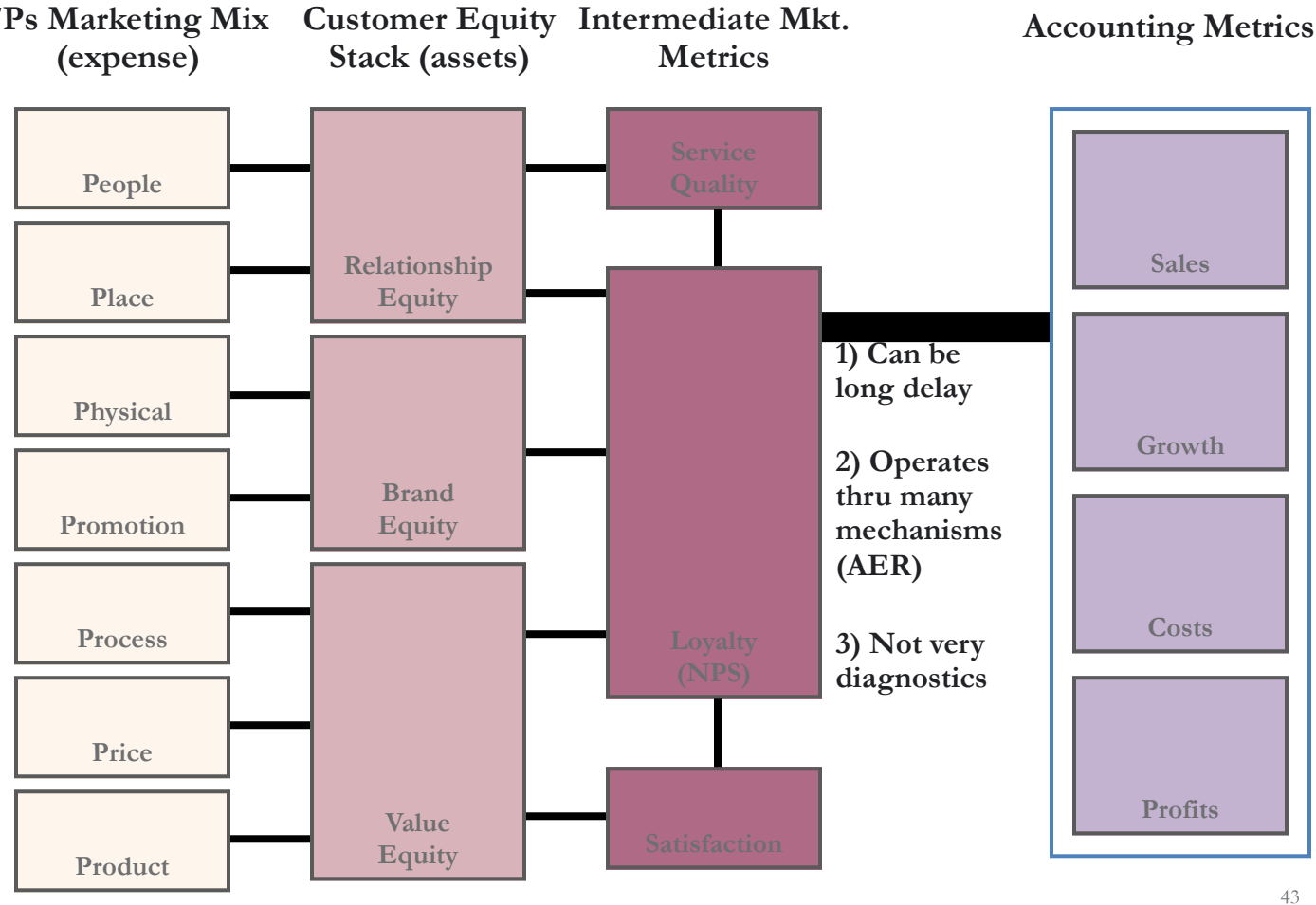
See *The Rules of Measurement and NPS* for overview

Benefits of Customer Loyalty

- Loyal customers tend to spend more with the organization over time (expansion, cross-selling)
- Less churn, higher retention
- Price premium
- On average costs of account maintenance are lower than for new customer costs
- Employee retention is more likely with a stable customer base (impact on firm culture)
- Positive WOM increases customer base and reduces acquisition costs
- Positive attributions and competitive sheltering
- Shares information and helps innovation
- Very strong positive affect on CLV

See *Building Loyalty in Business Markets*

Loyalty is Better than Accounting Metrics, but…



Many Marketing Metrics, But Two Main Approaches: Pros and Cons?

- Use **specific metric** for what you want to control (narrow, fast, and domain specific)
  - → Brand (awareness, quality, strength)
  - → Offering (innovativeness, adoption rates, price premium, value)
  - → Relationship (loyalty to salesperson, trust, relationship quality, velocity)
  - → Acquisition (# new customers, CLV, duration)
  - → Retention (churn, lost customer, share)
- Use **aggregate financial metrics** and analyze using chain ratios (uses accounting measures, can be decomposed)
  - → **Net marketing contribution** (NMC) tells us profit after sales and marketing costs
  - → **Marketing ROS** tells us what portion of our sales is profit (MROS =NMC/sales * 100%)
  - → **Marketing ROI** tells us our earnings for every 1$ invested in sales and marketing (MROI=NMC/Sales and marketing costs *100%)
  - ⇉

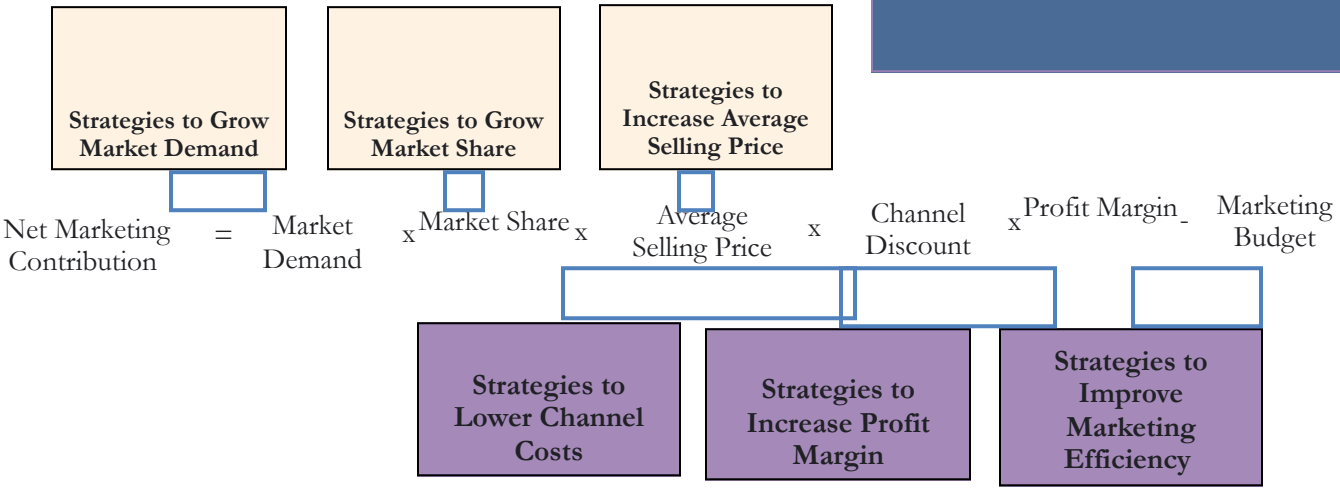
Types of Metrics

Net Marketing Contribution (NMC) as a Chain Ratio

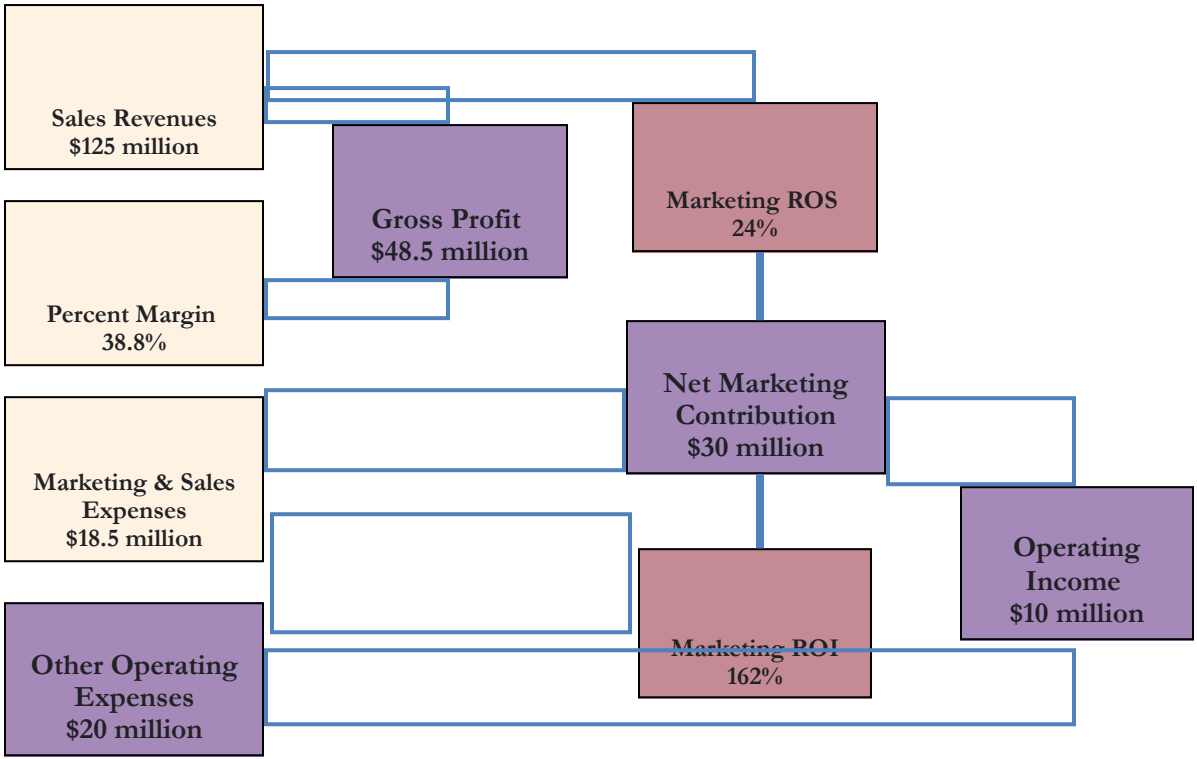
Sales
(COGS)
Gross Profit
(SGA)
Net Income

Sales
(COGS)
Gross Profit
(Sales and Marketing Costs)
Net Marketing Contribution
(GA)
Net Income

See *Note on Marketing Performance Assessment* for excellent overview of metrics



MROS & MROI Can be Used to Benchmark Entities (competitors, products, regions)



(Best 2012)

Effective Marketing Dashboards Capture Multiple Views of the Firm Over Time

Internal View of Performance

External View of Performance

(Best 2012)